Tracing Agentic Workloads at Scale

ploma demo

Abstract

LLM agents issue long chains of tool calls, and current tracing stacks record each call without the decision that caused it. We present Hawser, a tracer that records the causal link from each model decision to the tool results it consumed. Across 1200 tasks on three agent frameworks, Hawser cuts median trace-query latency by 27.4% against an OpenTelemetry pipeline. It stores 1.1 times the bytes of that pipeline.

1 Introduction

LLM agents plan, call tools and act on the results, often thousands of times within one task [7, 5]. When an agent fails, the operator needs to know which decision led to which tool call. The operator also needs to know which result changed the next decision. Request tracing records the path of a request through a distributed system [6, 1]. It treats each tool call as an independent span, so it loses the causal chain between a decision and the results it consumed. We present Hawser, a tracer that keeps this chain. Against an OpenTelemetry pipeline, Hawser cuts median trace-query latency by 27.4%.

This paper makes three contributions. First, a causal trace model that links each model decision to the tool results it read (Section 2). Second, a query planner that answers causal questions without scanning whole traces. Third, an evaluation on 1200 tasks across three agent frameworks (Section 3).

2 Method

Causal trace model. Hawser records two kinds of event. A *decision* is one model call together with the context it received. A *tool result* is the output of one tool call. Each decision carries edges to the tool results that appeared in its context, and each tool call carries an edge to the decision

Table 1: Reduction in median trace-query latency against each baseline, and Hawser storage as a multiple of the baseline's, 1200 tasks across three agent frameworks. Storage was measured against the OpenTelemetry pipeline only.

Baseline	Tasks	Latency cut (%)	Storage ratio
OpenTelemetry pipeline	1200	27.4	1.1
Raw logs	1200	41.2	n/a

that issued it. The trace of a task is therefore a directed acyclic graph rather than a tree of spans. Like Pivot Tracing [4], Hawser propagates a small amount of metadata with each call so that edges can be built at the point of use rather than reconstructed afterwards.

Query planner. Operators ask causal questions, such as which tool results influenced a failed decision. A naive answer scans the full trace and joins events by identifier. Hawser instead indexes each edge by its source and destination and walks the graph backwards from the decision of interest. The planner prunes branches whose events fall outside the requested time window. To bound storage, Hawser samples whole traces rather than single events, following work on biased trace sampling [2, 3].

3 Results

We ran 1200 tasks on three agent frameworks and issued the same set of causal queries against each tracing back end. Table 1 reports the reduction in median trace-query latency that Hawser achieves against two baselines, and its storage relative to the OpenTelemetry pipeline. Against an OpenTelemetry pipeline, Hawser cuts median latency by 27.4%. Against raw logs searched by identifier, it cuts median latency by 41.2%. Hawser stores 1.1 times the bytes of the OpenTelemetry pipeline, because it keeps the causal edges as well as the events.

References

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